CHEMWORLD | ARXIV V1 PUBLICATION PROOF

Executable Chemical Worlds for Measuring Experimental Agency

An executable, chemistry-native apparatus that separates experimental success from reproducible experimental behavior.

P0 release proof | 2 August 2026 | Frozen evidence and claim audit

29,754	15	28	5
executed physical experiments	registered task designs	typed operation types	instrument types

ABSTRACT

A successful experiment does not by itself show how an artificial-intelligence agent arrived there, whether it used intermediate evidence, whether it retained the discovered condition, or whether the same behavior would recur. Existing evaluations usually observe either an endpoint prediction, an optimized condition, or physical execution in a single laboratory system. We introduce ChemWorld, an executable chemical-world apparatus that makes the process of experimentation measurable. Agents act through typed, state-changing operations; choose when and what to measure; spend materials, vessels, instrument uses and operations from a campaign ledger; encounter explicit failures; and decide when to terminate and request a final assay. Hidden physical identity, material information and fresh agent trajectories can be controlled independently, while every accepted transition is identity-bound and physically replayable.

Compiled controls comprising 29,580 simulator executions, organized around ten paired physical worlds per task and information arm, show that task outcome, held-out prediction, calibration and unsupported claims are non-interchangeable readouts. Under primitive control, native Codex completed 60 of 60 autonomous electrochemical experiments through 815 self-selected operations. Similar endpoints arose through visibly different sequences of discovery, loss, retention and recovery. A commit-frozen fresh-session replication then completed 18 of 20 cells; two cells were right-censored by provider-infrastructure failure, leaving eight complete within-world pairs. Six of eight pre-specified world-by-lifecycle classifications were mixed, and at least six remained mixed under every possible sign of the two missing pair differences. The result was also stable across 80%, 90% and 95% retention definitions. ChemWorld therefore turns experimental agency from an endpoint score into a controlled object of study: experimental success can be distinguished from reproducible experimental behavior.

Environment counts qualify the release surface; formal agent evidence covers the tasks stated in the manuscript.

1. Introduction

When a scientific agent reports a high-performing condition, the most important question is no longer only *what score did it obtain?* It is also *what kind of experimental process produced that score?* A condition may be found early and retained, found once and abandoned, recovered after a drawdown, or reached at the end of an otherwise uninformative search. These trajectories imply different capabilities even when their final values agree.

This distinction is difficult to measure with existing evaluation regimes. Prediction benchmarks remove the act of experimentation. Optimization benchmarks expose a value-returning oracle but usually collapse each experiment to a vector query. Self-driving laboratories and instrument-operating agents provide indispensable evidence about physical execution, automation and deployment [@szymanski2023alab; @dai2024mobile; @darvish2025organa; @vriza2026instruments], but a physical apparatus cannot usually be cloned many times while changing only the information supplied to the agent. As a result, the agent's experimental behavior is often entangled with a particular laboratory, material batch and observation history.

ChemWorld fills this measurement gap with executable chemical and chemical-engineering worlds. The central design choice is simple: **the agent is the experimental subject; the chemical world is the apparatus**. Within a stateful, partially observable world, the agent selects additions, process conditions, measurements, termination and final assay. The environment records the consequence of every choice, including invalid operations, failed attempts and resource use. Matched hidden-world identities permit controlled information contrasts, while fresh sessions in the same physical world measure behavioral repeatability.

This apparatus changes what can be asked of a scientific agent. Instead of reducing performance to one leaderboard value, we can measure lifecycle completion, within-campaign best-discovery position, retention of an incumbent, drawdown, recovery, terminal quality, prediction and claim reliability. These are operational readouts of behavior, not inferred mental states, and they need not move together.

We use three evidence layers. First, compiled experiments provide a low-authority calibration in two task families and show that information changes actions and outcomes in task-dependent ways. Second, primitive-control campaigns show that a general agent can close complete experimental lifecycles under a resource ledger. Third, fresh trajectories in matched physical worlds test whether an observed information response reappears. Together they establish the paper's central result: **an experimental endpoint is insufficient evidence for a reproducible experimental strategy**.

Our contributions are:

1. an executable, chemistry-native apparatus with stateful operations, active measurements, failures, resource accounting, identity control and exact physical replay;
2. a primitive-control protocol in which the agent---rather than a fixed recipe---chooses operations, observations and lifecycle termination;
3. trajectory-level measures that separate discovery, retention, drawdown, recovery and terminal behavior; and
4. a matched-world fresh-session design showing how to test the repeatability of an agent's experimental behavior rather than only its endpoint.

2. Relation to existing systems

Self-driving laboratories optimize reactions and materials in real hardware, and chemistry agents increasingly connect language models to literature, planning tools and robotic execution [@boiko2023autonomous; @bran2024augmenting; @panapitiya2026autolabs; @pilon2026robochemflex]. Their defining strength is physical validity: they answer whether a proposed workflow can be executed and whether automation can operate reliably. ChemWorld addresses a complementary question---whether the experimenting agent can be isolated, intervened on and repeated under matched physical identity.

Digital optimization suites provide controlled objective functions and scalable algorithm comparison [@felton2021summit; @hase2021olympus]. Interactive science environments extend this idea toward stateful control, discovery and model identification [@beeler2024chemgymrl; @bloor2024pcgym; @jansen2024discoveryworld; @gandhi2025boxinggym; @duan2025scigym; @nagele2026sciexplorer]. ChemWorld's distinctive unit is a chemical experiment as a resource-consuming sequence of typed operations and measurements, not a single parameter query. The same world can be replayed under different information and authority conditions without changing its hidden physical identity.

Recent work treats agents themselves as behavioral objects and studies how environment design changes scientific discovery [@chen2026agentbehavior; @xin2026eurekagent; @riosgarcia2026scientifically]. ChemWorld brings that perspective to chemistry-native experimental lifecycles. Its contribution is the combination of matched chemical worlds, primitive action authority, evidence access, campaign-wide resources, immutable trajectories and within-world fresh-session replication. It does not replace a robotic laboratory; it provides the controlled measurement apparatus that a real laboratory cannot economically instantiate at scale.

3. ChemWorld makes experimental agency measurable

3.1 Executable worlds rather than value-returning oracles

A ChemWorld task defines a hidden physical state, public observation contract, typed operations, instrument responses, resource endowment, failure semantics and one or more evaluator-bound endpoints. Operations change state. Measurements are actions with costs and timing consequences. Termination closes a vessel, and a final assay is explicitly requested rather than automatically revealed.

The qualified environment surface contains 15 registered tasks, 28 operation types, five instrument classes, 415 deterministic complete-experiment boundary cases and 62 evaluator-bound endpoints (Fig. 1). Qualification establishes runtime reachability and evaluator binding. The empirical paper uses two tasks for compiled controls and one for autonomous primitive control, keeping platform scope and evidence scope explicit.

3.2 Controlled identity, resources and replay

Every run binds source, configuration, task, physical world, material instance, observation namespace and trajectory identity. The public agent view excludes hidden evaluator state. Accepted operations are transactional: a successful transition updates state and consumes the corresponding resources; a rejected operation records its failure without silently repairing the action.

The campaign ledger spans raw materials, solvent, vessels, non-final instrument uses, operation attempts and provider sessions. This design makes opportunity cost part of the experimental record while keeping the ledger external to the prompt. Agents access compact status and history artifacts on demand rather than receiving a continually expanding transcript.

Exact replay means exact reconstruction of environment transitions, public observations and resource changes from the stored trajectory. It does not mean regeneration of the language model's token sequence. Fresh-session replication therefore deliberately samples a new agent trajectory while holding the physical world fixed.

3.3 Evidence layers and analysis units

The number of executed world transitions measures use of the apparatus, not the number of statistically independent samples. Table 1 separates those quantities throughout the paper.

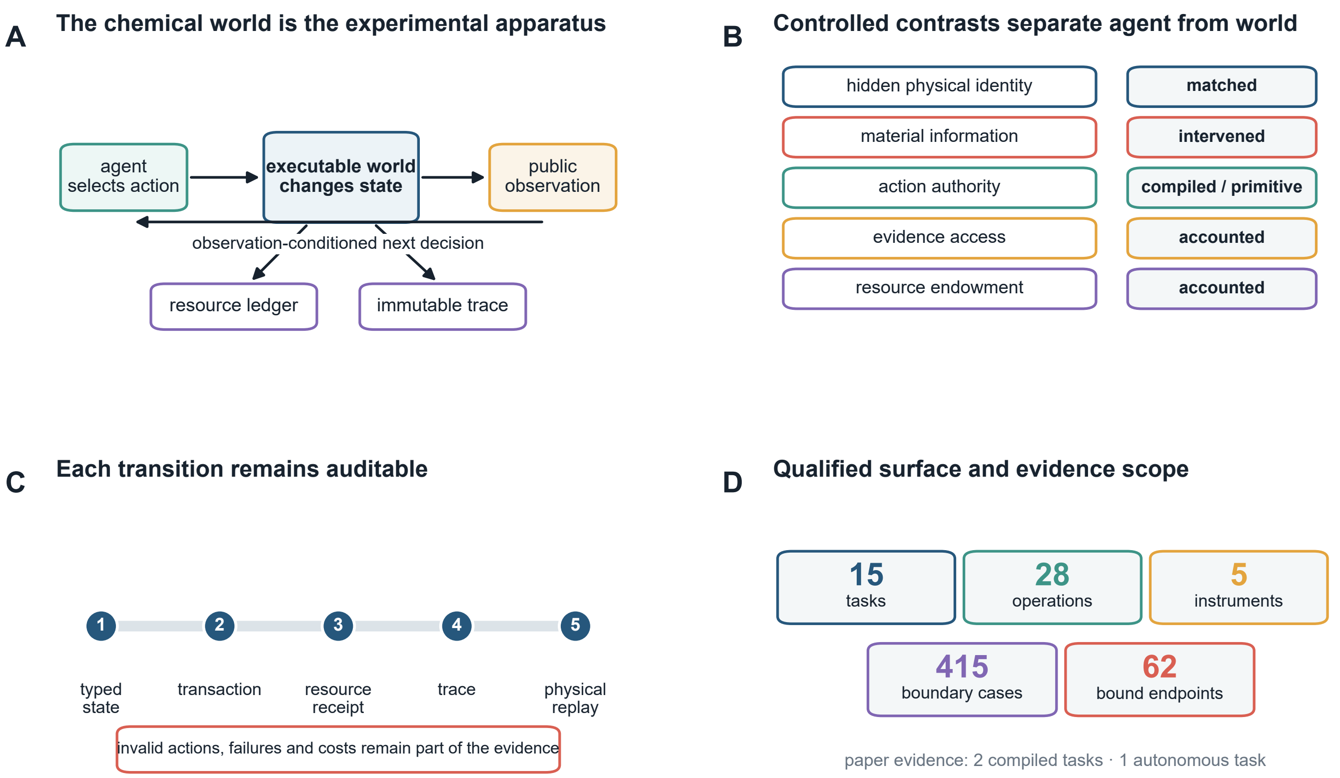

Figure 1 | ChemWorld is a controlled apparatus for measuring experimental agency. Typed state transitions couple public observations to resource receipts, immutable traces and exact physical replay while experimental contrasts remain explicit.

Table 1 | Qualified environment surface and formal evidence scope

Quantity	Count	Evidence level
registered task designs	15	release surface
typed operation types	28	release surface
instrument types	5	release surface
deterministic complete-experiment cases	415	design qualification
declared endpoints bound to evaluators	62	design qualification
tasks with formal compiled-agent results	2	paper evidence
tasks with autonomous-agent results	1	paper evidence

Counts for the environment surface are design qualifications, not claims of agent competence. Formal paper evidence covers fewer tasks than the registered surface.

4. Compiled controls separate outcome from evidence use

Compiled control gives the agent a bounded complete-experiment interface. It is not the target autonomy setting; it is a controlled calibration that can be run across tasks, information conditions and classical optimization policies.

Across ten paired worlds, the nominal-information arm produced a mean nominal-minus-opaque score difference of 0.072 in electrochemical conversion (8/10 positive worlds; familywise 97.5% world-bootstrap interval 0.007 to 0.155) and 0.026 in reaction-to-crystallization (7/10 positive; interval -0.013 to 0.063). The contrast is therefore task-conditioned rather than a universal benefit of material information (Fig. 2A).

The deliberately misindexed arm provides a manipulation check. It redirected the first material-sensitive action in 70% of electrochemical worlds and 100% of crystallization worlds. Misleading-action share subsequently fell from 0.54 to 0.24 in electrochemistry and from 0.86 to 0.50 in crystallization (Fig. 2B). The distinction among action manipulation, later correction and performance restoration is important: both tasks passed the behavior-change check, while their correction and score-restoration checks differed, and neither passed the joint rule (Fig. 2C).

Outcome and epistemic diagnostics also gave different profiles. In the opaque arm, electrochemical conversion combined a 0.715 endpoint score with 0.744 held-out directional accuracy, a 0.186 Brier score and a 0.611 unsupported-claim rate. Reaction-to-crystallization yielded 0.535, 0.478, 0.298 and 0.714, respectively. These measurements are reported separately because none is a substitute for another (Fig. 2D).

Classical optimizers are retained as calibration controls rather than the paper's target competition. Complete world-level contrasts for every baseline, including exact sign summaries and leave-one-world-out ranges, are provided in the sensitivity artifact. This keeps the empirical question aligned with the paper's contribution: what aspects of experimental behavior become measurable once the world is a controlled apparatus?

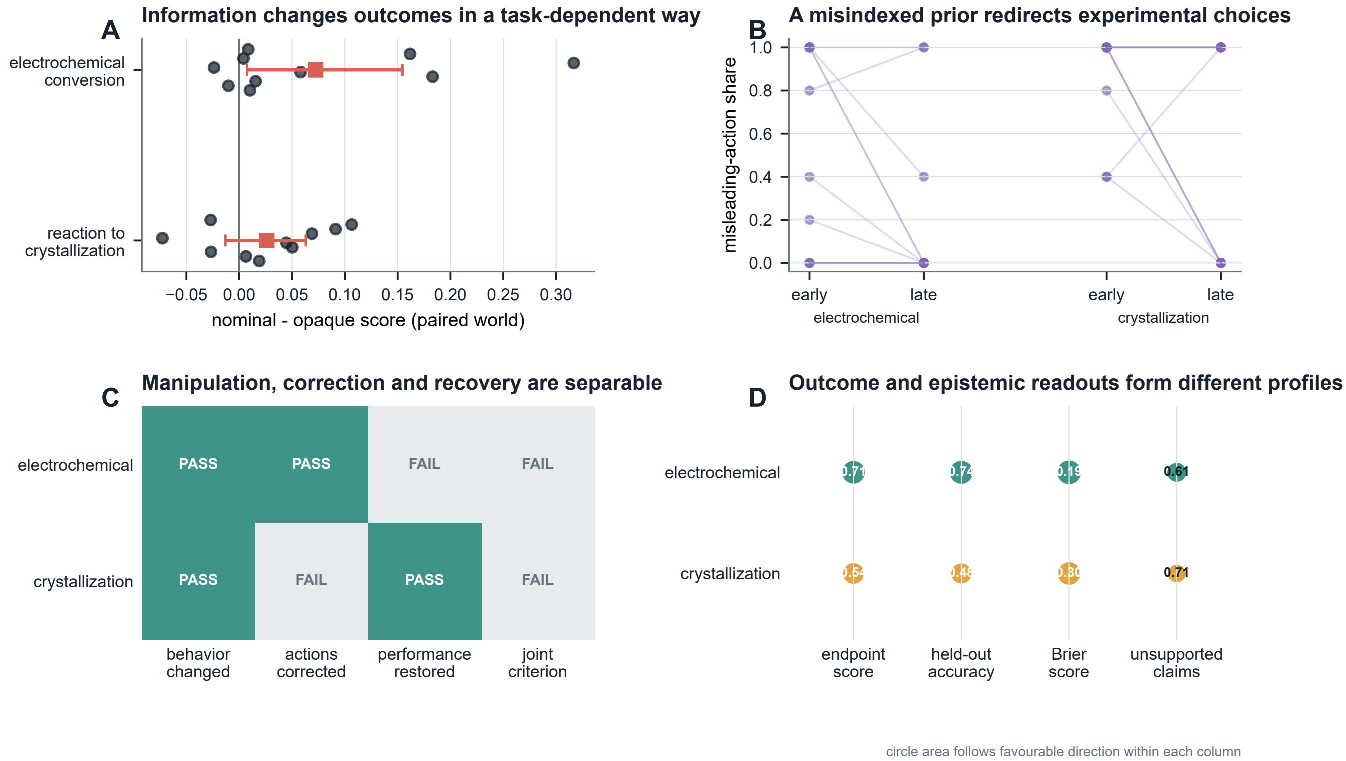

Figure 2 | Compiled controls distinguish task outcome, information response and epistemic readouts. Paired worlds, a misindexed-prior manipulation and separate outcome and epistemic measures establish the low-authority calibration layer.

Table 2 | Compiled-experiment capability profiles

Task	Information arm	Worlds	Final score	Held-out accuracy	Brier	Structure F1	Mechanism F1	Unsupported claims
electrochemical-conversion	opaque	10	0.715	0.744	0.186	0.389	0.190	0.611
electrochemical-conversion	nominal	10	0.787	0.778	0.149	not measured	not measured	not measured
electrochemical-conversion	misindexed	10	0.685	0.711	0.209	not measured	not measured	not measured
reaction-to-crystallization	opaque	10	0.535	0.478	0.298	0.275	0.144	0.714
reaction-to-crystallization	nominal	10	0.562	0.433	0.316	not measured	not measured	not measured
reaction-to-crystallization	misindexed	10	0.584	0.433	0.316	not measured	not measured	not measured

Scores are means across ten physical worlds. Dashes indicate endpoints that were not defined for that information arm; they are not zeroes. No composite score is formed.

5. Primitive control closes full experimental lifecycles

In a separate primitive-control study, native Codex, configured as `gpt-5.6-sol` at medium reasoning effort, received a public workspace and typed tools. For each vessel it selected material additions, process setpoints, electrolysis operations, measurements, termination and final assay. The model could inspect status and history artifacts, but the complete resource ledger was not inserted into the prompt.

All ten development cells completed all six vessels: 60 complete autonomous experiments through 815 accepted primitive operations (Fig. 3). The agent used 18 non-final measurements per campaign at most, while every campaign had a 144-operation ceiling and fixed stocks of reagents and solvents. Physical-pair, resource-ledger and exact-transition replay audits passed for all cells.

Figure 3A reconstructs one vessel. The important feature is not the particular seven-operation sequence; it is the placement of observation before the next agent decision. A UV--visible measurement entered the public state before the agent chose to terminate and request the final assay. Figure 3B then shows that this lifecycle closed across every vessel and information arm, while Fig. 3C reconstructs the campaign-level resource receipt.

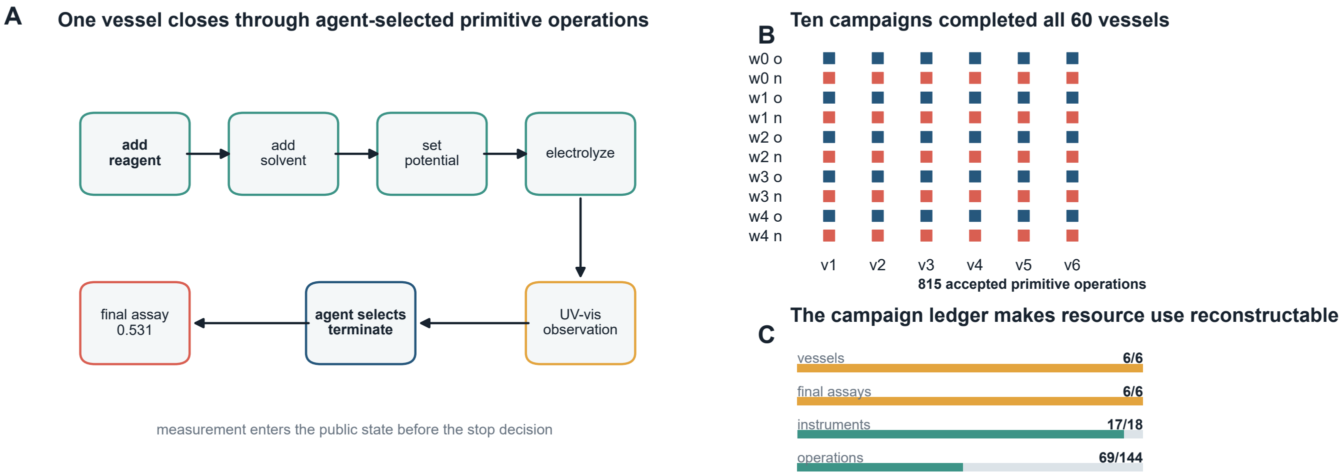

Figure 3 | Primitive-control agents close complete experimental lifecycles. The agent selects operations, observes intermediate evidence, decides when to stop and requests final assay under a reconstructable campaign ledger.

Table 3 | Autonomous development trajectories (G2 v0.4)

Arm	Cells	Completion	Operations	Best score	Batch AUC	Realized-op AUC	Fixed-budget AUC	Discovery fraction	Retention	Max drawdown	Terminal / best	Recovery
opaque	5	1.000	82.600	0.631	0.617	0.520	0.563	0.320	0.520	0.333	0.671	0.500
nominal	5	1.000	80.400	0.709	0.589	0.501	0.591	0.800	0.720	0.092	0.941	0.800

Each arm contains five physical-world cells and six completed vessels per cell. Operations are mean submitted primitive attempts per cell. These development data select the worlds and endpoints for G2 v0.5 and are excluded from its replication estimand.

6. Endpoint scores conceal experimental trajectories

Primitive control exposes structure that a final recommendation cannot contain. Figure 4 follows six final assays in three development worlds. In world 0, the opaque trajectory lost its initial score, recovered by the fourth assay and finished near the nominal trajectory. In world 2, the opaque arm discovered its campaign best immediately and then lost it, whereas the nominal arm improved and retained a higher terminal value. In world 4, both arms began similarly but their terminal behavior diverged sharply.

We operationalize these patterns with four lifecycle readouts. Within-campaign best-discovery position records when the observed campaign maximum first appears. Online retention asks whether a new assay remains within a pre-specified fraction of the pre-assay incumbent. Maximum absolute drawdown records the largest departure from that incumbent. Terminal-to-best ratio records how much of the observed campaign best remains at the final assay. Loss episodes and recovery delays provide additional summaries when a trajectory falls below its retention boundary.

Across the five development worlds, nominal information was associated with later within-campaign best discovery on average (0.80 versus 0.32 of the six-assay horizon), together with higher retention (0.72 versus 0.52), smaller mean maximum drawdown (0.092 versus 0.333), greater pooled recovery (0.80 versus 0.50) and higher terminal-to-best ratio (0.941 versus 0.671). Because each arm contains one session per development world, these values define hypotheses for fresh-session replication rather than a population effect.

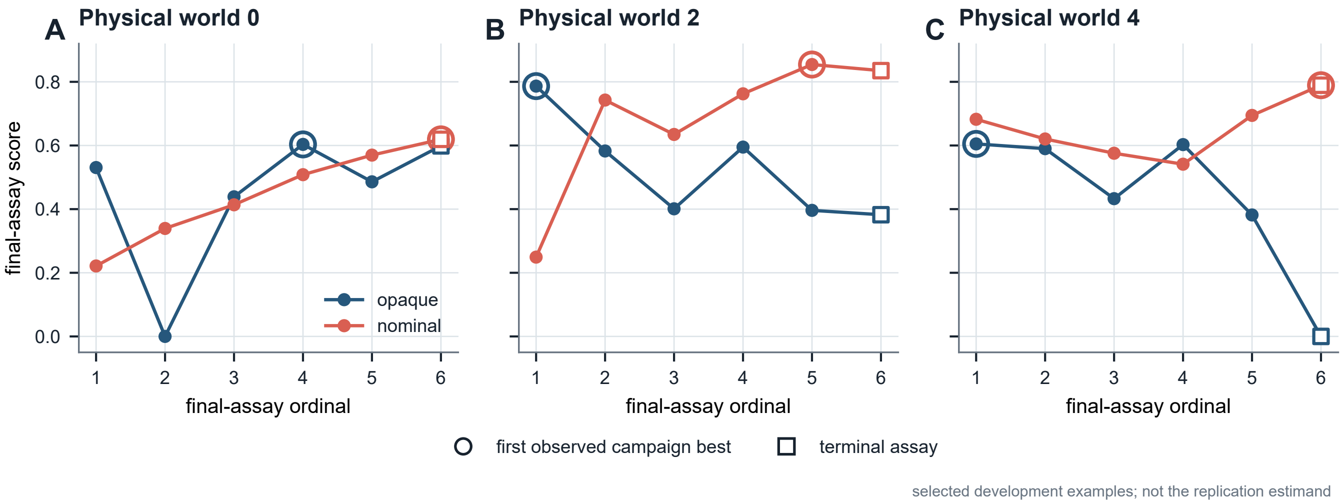

Figure 4 | Similar endpoints can arise from different experimental trajectories. Development worlds expose early discovery, loss, gradual improvement, retention and terminal divergence that an endpoint alone cannot resolve.

7. Fresh sessions test within-world repeatability

We selected physical worlds 1 and 3 before the replication launch because their development contrasts pointed in different directions. The commit-frozen design crossed two worlds, five fresh trajectory replicates and two information arms. Each cell received six vessel opportunities and the same physical, observation and resource identities as its within-replicate counterpart; the provider did not expose a controllable sampling seed.

Eighteen of 20 cells completed. Two cells were right-censored after 50 and 56 accepted operations by provider-infrastructure failures, leaving four complete pairs in each world. All ten planned pairs remain visible in Fig. 5. The pre-specified rule classified a world-by-metric result as directional when at least three of four available differences shared a sign and the median shared that direction.

The authoritative matrix followed a launcher-level restart after an infrastructure incident in the first launch; the restart decision was recorded before outcome inspection. The protocol deviation, excluded trajectories and separate sensitivity check are reported in Methods 10.7.

Endpoint direction alone did not determine lifecycle direction. World 1 was nominal-favouring for best score, smaller drawdown and terminal-to-best ratio, while discovery and retention were mixed. World 3 was mixed for all five displayed readouts. Across the four pre-specified lifecycle metrics, six of eight world-by-metric classifications were mixed.

This result is robust to the two missing pair differences: assigning each missing difference an arbitrary positive, negative or zero sign leaves at least six of eight lifecycle classifications mixed. At the 90% retention definition, six of eight were mixed; the count was five at 80% and six at 95%. Requiring four of four matching signs (an 80% directional threshold with four complete pairs) classified all eight as mixed. Across the full threshold-by-zero-handling grid, the count ranged from two to eight; the minimum occurred at a 60% threshold when exact zeros were excluded. Thus the pre-specified result and its missing-sign robustness are preserved, while the broader frequency of mixed labels depends on how exact-zero contrasts are treated.

The empirical statement is deliberately precise: the opposing development phenotypes did not become stable directional lifecycle contrasts across the available fresh sessions in these two physical worlds. That observation is exactly what matched executable worlds make visible and testable.

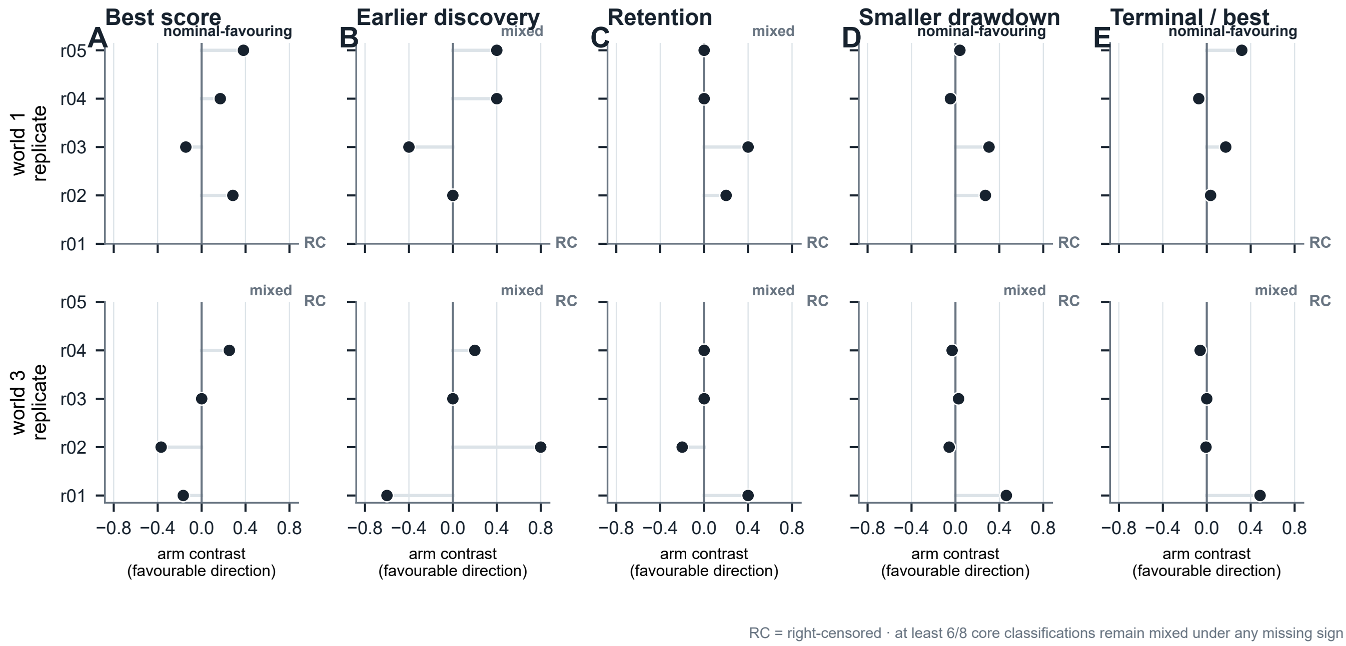

Figure 5 | Fresh trajectories separate endpoint direction from lifecycle repeatability. Matched-world session pairs keep right-censored cells visible and show mixed lifecycle directions under every possible sign of the missing differences.

Table 4 | Fresh-trajectory replication (G2 v0.5)

World	Replicate	Opaque state	Nominal state	Δ best score	Δ mean score	Δ discovery	Δ retention	Δ drawdown	Δ terminal / best
1	r01	completed	right_censored	not measured	not measured	not measured	not measured	not measured	not measured
1	r02	completed	completed	0.285	0.304	0.000	0.200	-0.273	0.035
1	r03	completed	completed	-0.143	-0.074	0.400	0.400	-0.306	0.173
1	r04	completed	completed	0.170	0.144	-0.400	0.000	0.045	-0.073
1	r05	completed	completed	0.381	0.429	-0.400	0.000	-0.040	0.319
3	r01	completed	completed	-0.167	-0.121	0.600	0.400	-0.463	0.486
3	r02	completed	completed	-0.368	-0.236	-0.800	-0.200	0.057	-0.007
3	r03	completed	completed	0.001	-0.009	0.000	0.000	-0.028	0.000
3	r04	completed	completed	0.253	0.291	-0.200	0.000	0.030	-0.060
3	r05	right_censored	completed	not measured	not measured	not measured	not measured	not measured	not measured

Deltas are nominal minus opaque within the same physical world and replicate block. The two deliberately selected worlds are not pooled into a population-level estimate. Terminal coverage: 18 completed cells, 2 right-censored cells, and 8 complete pairs (4 in world 1; 4 in world 3). The frozen interpretation mapping selected `frequent_within_world_reversal`: 6 of 8 world-by-core-lifecycle classifications were mixed. Policy SHA-256: `93604ce8af7211f35c5d3b896609addef6d436f3248f45aba3cc04a11da9d67e`. Frozen descriptive summary: the available fresh-session contrasts frequently changed direction within the selected physical worlds. Provider sampling was not seed-controlled; the summary does not identify a causal provider effect or a variance-dominance relation.

8. Experimental agency is a profile, not a scalar

The compiled and primitive-control studies measure complementary parts of experimental agency. Compiled control offers repeated task and information contrasts together with prediction, calibration and claim diagnostics. Primitive control reveals lifecycle completion, retention, recovery and terminal behavior. Figure 6 places these readouts side by side without combining them into a composite score.

The profile view supports a stronger evaluation standard for scientific agents. Success should be accompanied by evidence that the agent can complete the experimental lifecycle, make observations available to later decisions, retain or recover a productive strategy and reproduce the relevant behavior in a fresh session. ChemWorld provides the controls needed to measure those properties separately.

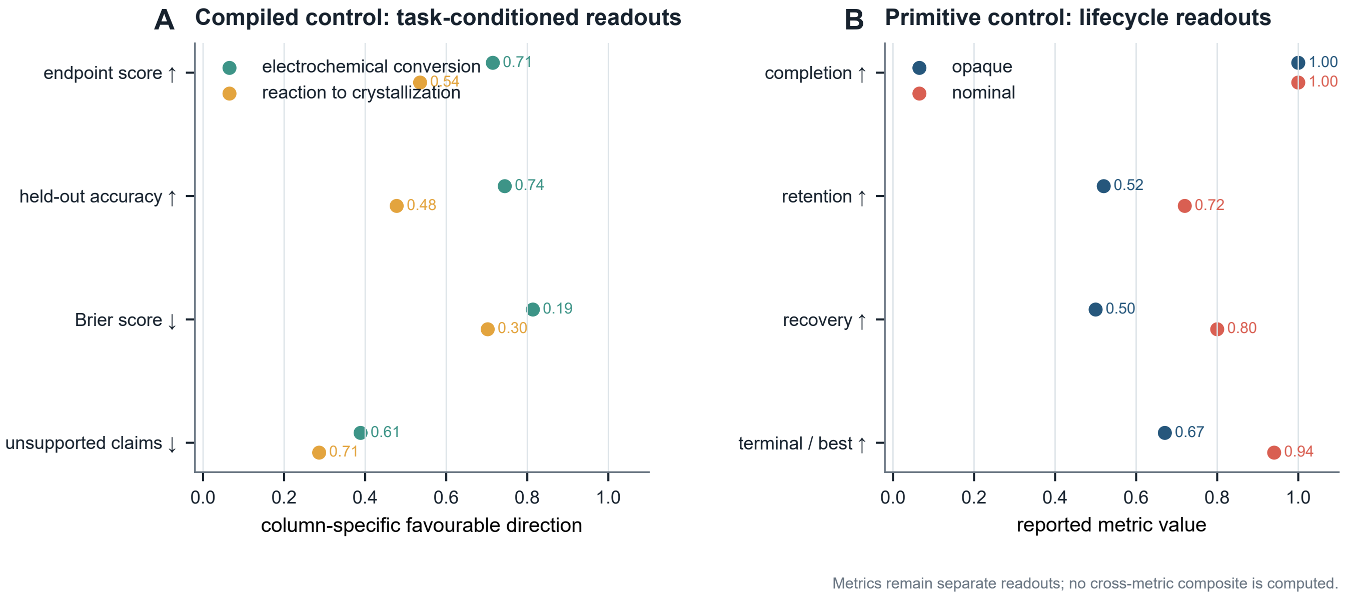

Figure 6 | Experimental agency is resolved as a profile of separate readouts. Outcome, prediction, calibration, claim reliability, completion, retention, recovery and terminal behavior remain separate rather than being collapsed into a composite score.

9. Discussion

ChemWorld contributes a new experimental object: the process by which an agent experiments. Its value does not depend on replacing a self-driving laboratory or winning a universal optimization leaderboard. It occupies a different and useful role. A real laboratory establishes physical executability; an optimization suite calibrates search; an executable chemical world can clone physical identity, intervene on information and authority, preserve failures, and sample fresh decision trajectories at low marginal cost.

The present experiments demonstrate three consequences. First, information can change actions without producing a uniform task-independent endpoint effect. Second, autonomous lifecycle completion does not specify how a high-performing condition was discovered, retained or recovered. Third, a directional endpoint contrast can coexist with mixed lifecycle contrasts across fresh sessions. The apparatus therefore distinguishes experimental success from reproducible experimental behavior.

The next scientific use of this apparatus is not simply a larger leaderboard. It is a factorial measurement program: randomly sampled worlds, multiple mechanism families, multiple agents, controlled evidence access and matched resource endowments. Such experiments can estimate world heterogeneity, trajectory variability and intervention response hierarchically, and can test whether lifecycle reliability predicts transfer, recovery from misinformation or behavior under scarcity.

Synthetic instruments and scoring laws make those interventions inexpensive and repeatable. A calibrated physical or high-fidelity bridge can then test which trajectory phenomena transfer to a particular laboratory system. The two forms of evidence are complementary: executable worlds identify controlled behavioral structure; physical laboratories establish deployment validity.

10. Methods

10.1 Environment qualification

The task registry and runtime-domain audit declare task contracts, typed operations, instrument interfaces, state invariants and evaluator-bound success metrics. Qualification required deterministic reachability through 415 complete boundary cases and binding of all 62 declared endpoints to evaluator state. The qualified surface contains 15 tasks, 28 operation types and five instruments. These counts characterize supported environment capabilities; empirical agent results are reported only for the task families actually exercised.

10.2 Compiled-control protocol

Compiled control covered electrochemical conversion and reaction-to-crystallization. Each task used physical world seeds 0--9 and three information conditions: opaque material codes, anonymous nominal properties and a commit-frozen misindexing intervention. Each participant session selected 20 complete experiments, after which a replay verifier recomputed all scores from the immutable reports. Classical policies used the same world identities, budgets and information contracts.

Compiled participants used the Codex subscription transport with model alias `gpt-5.6-sol` at medium reasoning effort, structured response output, Codex tools disabled and no session persistence. G0 and G2 therefore share a model alias and reasoning setting but use different scaffolds and action interfaces; they are complementary evidence layers, not a matched causal comparison of authority.

The nonduplicated total comprises 2,280 participant executions and 27,300 classical-control executions. Statistical summaries treat the paired physical world, not each simulator execution, as the analysis unit. Information contrasts report mean and median world differences, positive/negative counts, exact sign summaries, leave-one-world-out mean ranges and commit-frozen world-bootstrap intervals. Classical comparisons remain descriptive and all policies are shown in the public sensitivity artifact.

The misindexing manipulation transposed one targeted material row selected on independent qualification worlds. Commit-frozen checks separately evaluated initial behavior change, later action correction, performance restoration and their joint criterion.

10.3 Primitive-control protocol and resource ledger

The autonomous electrochemical protocol exposed typed tools for adding reagent and solvent, setting potential/current/material profile, electrolyzing, measuring, inspecting public status/history, terminating and requesting the final assay. Each cell contained six vessel opportunities. The campaign card bounded vessels, raw stocks, non-final instrument uses, operation attempts and provider sessions. Resource state was stored in an external artifact and exposed through compact queries; it was not repeatedly copied into the model context.

Native Codex used model alias `gpt-5.6-sol` with medium reasoning effort. Each vessel used a fresh provider session. The environment accepted only typed tool transactions; explanatory prose could not change physical state. A cell was complete only when every vessel was terminated and had a committed final assay.

10.4 Operational trajectory readouts

Let $s_1,\ldots,s_K$ be the final-assay scores in a campaign, with $K=6$, and let $b_t=\max_{j\leq t}s_j$ be the online incumbent.

Within-campaign best-discovery position is the first index of the observed campaign maximum, normalized to $[0,1]$:

For retention fraction q, assay $t>1$ retains the incumbent when $s_t\geq q b_{t-1}$. Online retention is the fraction of the $K-1$ opportunities satisfying this rule. A loss episode begins below the same boundary; its reference incumbent remains fixed until a later assay recovers to that boundary. Maximum absolute drawdown is $\max_t\max(0,b_{t-1}-s_t)$. Terminal-to-best ratio is $s_K/\max_t s_t$ when the observed best is positive. The primary retention fraction was $q=0.90$; $q=0.80$ and $q=0.95$ were sensitivity definitions.

These are operational trajectory readouts. “Discovery” refers to discovery of the best condition observed within that campaign, not identification of the global optimum of the hidden world.

10.5 Fresh-session replication

The replication crossed physical world seeds 1 and 3, trajectory replicates `r01`--`r05`, and opaque/nominal information conditions. Worlds were selected from the development set before launch because their observed development directions differed. Development trajectories were excluded from the replication estimand. Pair order and within-pair arm order were frozen in the schedule.

A provider failure before any accepted operation could be retried up to the frozen limit. A provider or method-resource failure after an accepted operation made the cell terminal and right-censored. Completed and right-censored cells in the authoritative launch were not replaced. Pair differences were computed only when both arms completed; no missing outcome was imputed.

For each world and metric, the primary descriptive rule classified a direction when at least 75% of available differences shared a sign and the median shared that direction. With four complete pairs this is a three-of-four rule. Otherwise the result was mixed. The four primary lifecycle metrics were best-discovery position, online retention, maximum absolute drawdown and terminal-to-best ratio; best and mean scores were endpoint diagnostics.

10.6 Sensitivity analyses

The frozen primary analysis was not changed. A separately hashed P0 sensitivity artifact evaluated directional thresholds 0.60, 0.75 and 0.80; inclusion or exclusion of exact zero differences; retention fractions 0.80, 0.90 and 0.95; and every positive/negative/zero sign assignment to the two missing pair differences. Because lifecycle metrics share the same six assay outcomes, the eight world-by-metric classifications are treated as a descriptive summary, not as eight independent inferential units.

10.7 First-launch infrastructure incident

The commit-frozen replication protocol was first launched on 1 August 2026. A detached outer Python process disappeared after `cell-001` had completed six vessels and `cell-002` had recorded four accepted operations. The cell-level protocol specified right-censoring after any accepted action and no rerun of a terminal cell. The decision to exclude and restart the entire launch therefore constitutes a protocol deviation at the launcher level.

The decision was made without changing the protocol, schedule, worlds, arms, budgets or analysis rule. The original directory was retained immutably; its completed cell and partial trajectory are included in the public trajectory archive. The authoritative matrix is the second launch and remains the primary analysis. The excluded launch is not pooled into primary pairs. As a transparent descriptive check, its completed nominal world-1/`r01` cell had six scores from 0.654 to 0.730 (mean 0.710); pairing it across launches with the corresponding formal opaque trajectory gives positive best-score, mean-score, retention and terminal contrasts and a smaller drawdown. This direction is compatible with, and not required for, the primary conclusion that at least six lifecycle classifications remain mixed.

10.8 Provenance, public boundary and replay

Source, configuration, world, material, observation and trajectory identities are SHA-256 bound. The evaluator trajectory contains hidden physical identity; the agent-facing `current.json` and `history.jsonl` expose only the public schema. Public-boundary tests compare those schemas and inspect representative workspaces for hidden-field leakage.

The release contains the self-hashed derived-data object, P0 sensitivity object, world-level tables, all 20 compact formal trajectories, both durable trajectories from the excluded first launch, a terminal file index and an independent verification attestation. Compact trajectories omit provider response content and hidden evaluator identity while retaining the fields required for exact physical-transition and resource replay.

11. Data and code availability

Code, configuration, derived data, figure generators, the arXiv source package and paper-sufficient public trajectories are publicly available in the MIT-licensed ChemWorld repository at github.com/sunyrain/ChemWorld/tree/arxiv-v1. The versioned paper release is rooted at `benchmark/releases/chemworld-serious-v1`, and its manifest binds every paper artifact to its SHA-256 identity. The G0 raw-file index covers 1,441 files across four immutable source roots; the tracked world-level summaries and public derived-data object reproduce every main-text number, table and figure. The G2 public archive contains compact replayable trajectories for all formal cells and for the first-launch infrastructure incident. Provider authentication, unrestricted provider responses and hidden evaluator identities are excluded from the public package. The 17.7-GB G0 raw roots are bound by the public file-level hash index but are not included in the repository and have not yet received a durable external archive identifier; raw-byte access is therefore not presently available from a permanent archive. The tracked world-level summaries and derived-data object are sufficient to regenerate the paper's reported analyses.

12. Conclusion

ChemWorld turns the act of experimentation into a controlled scientific object. Its executable worlds allow an agent to choose operations, seek evidence, spend resources, experience failure and close experimental lifecycles while physical identity remains matched and every consequence remains replayable. The resulting evidence separates an endpoint from the process that produced it: task outcome, prediction, retention, recovery and fresh-session repeatability form distinct readouts. The central message is therefore simple and actionable: **a successful experiment is a result; reproducible experimental agency is a stronger capability, and it can now be measured.**

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